🐳 Running AI Locally with Docker Desktop

A step-by-step guide to replacing GitHub Copilot with a free, private, GPU-accelerated local AI

🧠 Why Run AI Locally?

Problem with Cloud AI	Local AI Solution
💸 Costly at scale ($10–30/million tokens)	✅ Completely free
🌐 Needs internet to work	✅ Works fully offline
🔒 Your code goes to third-party servers	✅ Everything stays on your machine
⚡ Latency depends on network	✅ GPU-accelerated, sub-300ms responses

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🖥️ System Requirements

> This guide is optimized for the following setup. Adjust model size based on your hardware.

Component	Spec	Role in AI Inference
💻 Laptop	Lenovo IdeaPad Pro 5 16IMH9	—
🧠 CPU	Intel Core Ultra 9 185H	Handles overflow inference
⚡ NPU	Intel Arc	Dedicated AI acceleration chip
🎮 GPU	6GB NVIDIA	Primary model inference (fast!)
💾 RAM	32GB	Loads model weights
🗄️ Storage	1TB SSD	Stores model files
🪟 OS	Windows 11 Home	—

📦 Step 1 — Install Docker Desktop

Make sure you have **Docker Desktop** installed with the **AI features** enabled.

> Docker Desktop version must support the **Models** and **MCP Toolkit** sidebar items.

Download from: https://www.docker.com/products/docker-desktop

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🤖 Step 2 — Understanding Model Selection

Before pulling a model, you need to understand three key concepts:

🔢 Parameters

> The "brain size" of an AI model. More parameters = smarter model, but also larger file size and more RAM needed.

```
7B  = 7 Billion parameters  → Good for most coding tasks ✅
30B = 30 Billion parameters → Needs 16GB+ VRAM
70B = 70 Billion parameters → Needs very high-end hardware
```

📦 Quantization

> Compression of the model. Like image compression — smaller file, slight quality trade-off.

```
F16  → Full precision, 16 bits/parameter → Largest, best quality
Q8_0 → 8 bits/parameter                  → Good balance
Q4_0 → 4 bits/parameter                  → ~4x smaller, great for most tasks ✅
Q2   → 2 bits/parameter                  → Smallest, noticeable quality loss
```

💡 RAM vs VRAM Loading Rule

```
Model fits in VRAM (6GB)?  → Loads on GPU  ⚡ Super fast!
Model too big for VRAM?    → Overflows to RAM 🐢 Slower
Model too big for RAM?     → ❌ Won't run at all
```

🏆 Step 3 — Pull the Right Model

Our Pick: `qwen2.5:7B-Q4_0`

Property	Value	Why It Matters
Parameters	7.62B	Smart enough for coding tasks
Quantization	Q4_0	4x compressed, great quality
Size	4.12 GiB	Fits perfectly in 6GB VRAM ✅
Architecture	qwen2	Optimized for code

How to Pull

1. Open **Docker Desktop**
2. Click **Models** in the left sidebar
3. Search for `qwen2.5`
4. Find `qwen2.5:7B-Q4_0` and click **Pull**

Show Image

> ⏳ This will download ~4GB. Make sure you have a good internet connection.

⚡ Step 4 — Enable GPU-Accelerated Inference

By default, Docker Model Runner uses CPU only. Enabling GPU support gives **10x faster** responses!

Enable TCP + GPU Support

1. Open **Docker Desktop → Settings → AI**
2. Check ✅ **Enable Docker Model Runner**
3. Check ✅ **Enable host-side TCP support**
    - Port: `12434` (default — leave as is)

4. Check ✅ **Enable GPU-backed inference**
5. Click **Apply**

```
⚠️ Note: GPU-backed inference downloads additional components
   to ~/.docker/bin/inference — this may take a few minutes.
```

Verify It's Working

Open your browser and navigate to:

```
http://localhost:12434/engines/v1/models
```

You should see a JSON response like:

```json
{
  "data": [
    {
      "id": "ai/qwen2.5:7B-Q4_0",
      "object": "model"
    }
  ]
}
```

✅ If you see this — your model is running and accessible via API!

Verify GPU Usage

Go to **Docker Desktop → Models → Requests tab** after sending a prompt.

```
Duration: ~273ms  ← GPU is working! ⚡
Duration: ~3000ms ← CPU only 🐢
```

🔌 Step 5 — Connect VS Code via Continue.dev

Docker Models exposes a **local OpenAI-compatible REST API** at:

```
http://localhost:12434/engines/v1
```

This means ANY tool that supports OpenAI's API can use your local model — just change the URL!

Install Continue.dev Extension

1. Open **VS Code**
2. Go to **Extensions** (`Ctrl + Shift + X`)
3. Search for `Continue`
4. Install **Continue - open-source AI code agent** (2.4M+ downloads)

Configure Continue.dev

Open the config file at:

```
C:\Users\<yourname>\.continue\config.yaml
```

Replace everything with:

```yaml
name: Local Config
version: 1.0.0
schema: v1
models:
  - name: Qwen2.5 Coder Local
    provider: openai
    model: ai/qwen2.5:7B-Q4_0
    apiBase: http://localhost:12434/engines/v1
    apiKey: docker
```

> 💡 **Why `provider: openai`?** Docker's API is OpenAI-compatible — it speaks the same language, just at a different address (localhost instead of the cloud)!

> 💡 **Why `apiKey: docker`?** It's just a placeholder — no real authentication is needed for localhost APIs.

Fix Windows PowerShell Execution Policy (if needed)

If you see script execution errors, run this in **PowerShell as Administrator**:

```powershell
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Reload and Test

1. Click **Reload** in the Continue panel (right sidebar in VS Code)
2. Select **Qwen2.5 Coder Local** from the model dropdown
3. Ask it something:

```
Can you explain what a Docker volume is?
```

✅ If it responds — **you're done!** You now have a free, private, GPU-accelerated AI coding assistant running 100% on your machine! 🎉

🪜 How It All Works Together

```
┌──────────────────────────────────────┐
│              YOUR MACHINE            │
│                                      │
│   ┌──────────────────┐               │
│   │  Docker Models   │  ← qwen2.5:7B-Q4_0
│   │  (AI Brain) 🧠   │    runs on your GPU
│   └──────────┬───────┘               │
│              │ exposes               │
│              ▼                       │
│   ┌──────────────────┐               │
│   │ localhost:12434  │  ← OpenAI-compatible
│   │  REST API  🔌    │    just like -p 8080
│   └──────────┬───────┘               │
│              │ connects to           │
│              ▼                       │
│   ┌──────────────────┐               │
│   │  VS Code         │  ← Continue.dev
│   │  (Your IDE) 💻   │    extension
│   └──────────────────┘               │
└──────────────────────────────────────┘

        No internet. No API costs. No data leaks.
```

🆚 Before vs After

	Before (GitHub Copilot)	After (Local AI)
🧠 AI Brain	GitHub's servers	Your GPU
💸 Cost	$10-19/month	**FREE**
🔒 Privacy	Code sent to GitHub	**Stays on your machine**
🌐 Internet	Required	**Not needed**
⚡ Speed	Depends on network	**~273ms on GPU**
🛠️ Customizable	Limited	**Full control**

🚀 Quick Reference

```bash
# Verify model API is running
curl http://localhost:12434/engines/v1/models

# Check Docker Model Runner status
# Docker Desktop → Models → Requests tab

# Config file location
# C:\Users\<yourname>\.continue\config.yaml
```

❓ Troubleshooting

Issue	Cause	Fix
`Connection error` in Continue.dev	TCP not enabled	Docker Desktop → Settings → AI → Enable host-side TCP
Slow responses (>2s)	GPU not enabled	Docker Desktop → Settings → AI → Enable GPU-backed inference

Issue	Cause	Fix
`npx` script error	PowerShell policy	Run `Set-ExecutionPolicy RemoteSigned` as Admin
Model not showing	Not pulled yet	Docker Desktop → Models → Pull qwen2.5:7B-Q4_0

📅 Guide created: March 2026 | 🐳 Docker Desktop with AI features (Beta)